

Labwork 2

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1 GPU Information

The GPU information was obtained using the Numba CUDA API.

- **Device name:** NVIDIA GeForce GTX 1650
- **Multiprocessor count:** 14
- **Memory size:** 3,895,132,160 bytes
- **Memory size:** approximately 3.63 GB
- **Compute capability:** 7.5

The GPU has 14 multiprocessors and approximately 3.63 GB of reported device memory. Its compute capability is 7.5, which corresponds to the NVIDIA Turing architecture.

2 GPU Detection Code

The following Python code was used to obtain the GPU information:

```
from numba import cuda

gpu = cuda.get_current_device()

free_memory, total_memory = cuda.current_context().get_memory_info()

print("Device name:", gpu.name)
print("Multiprocessor count:", gpu.MULTIPROCESSOR_COUNT)
print("Memory size:", total_memory, "bytes")
print("Memory size:", total_memory / (1024**3), "GB")
print("Compute capability:", gpu.compute_capability)
```

3 Result

The program successfully detected the NVIDIA GeForce GTX 1650 GPU. The device contains 14 multiprocessors and reports approximately 3.63 GB of device memory. The reported compute capability is 7.5.